
MCA/MSc.(CS) Semester-II 2024-25
CS-322: Deep Learning

GROUP ASSIGNMENT

Max. Marks: 10

- (Q1.) Discuss and explain, in your own words, the types of errors, the bias-variance tradeoff, overfitting and underfitting, variants of gradient descent, and momentum. Support your explanation with appropriate references and diagrams.
- (Q2.) Discuss and explain the concept of Optimization in your own words, including appropriate references and relevant diagrams.
- (Q3.) Using vector calculus, derive the update rule for a specific variant of gradient descent. Provide your own example to illustrate the process.
- (Q4.) Discuss and explain Logistic Regression, covering the fundamental concepts of regression and classification problems, the use of linear models for both tasks, the principle of maximum likelihood, and the functioning of logistic regression classifiers.
- (Q5.) Implement logistic regression on a given dataset, perform data preprocessing (such as normalization or handling missing values), train the model, and evaluate its classification accuracy. Additionally, analyze the model's performance using metrics such as precision, recall, F1-score, and the confusion matrix. Explore the impact of hyperparameter tuning (e.g., learning rate, regularization) on model performance and provide a detailed report on the results. Provide your own example to illustrate the process.
- (Q6.) Discuss and explain the fundamental concepts of artificial neurons, single and multilayer perceptrons, the perceptron learning algorithm, various activation functions, and the softmax cross-entropy loss function.
- (Q7.) Discuss and explain Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) architectures, and the fundamentals of word embeddings.
- (Q8.) Discuss and explain Deep Reinforcement Learning and Autoencoders (including standard, denoising, contractive, etc.) in your own words, supported by appropriate references and illustrative diagrams.
- (Q9.) Discuss and explain the fundamental concepts of Convolutional Neural Networks (CNNs), beginning with the concept of filtering. Provide a detailed explanation of the convolution operation, including how filters interact with input data, and describe various types of filters used in CNNs. Elaborate on the pooling operation—such as max pooling and average pooling—and explain their role in reducing spatial dimensions and controlling overfitting. Additionally, illustrate the arithmetic involved in both convolution and pooling operations, including stride, padding, and output dimension calculation. Use examples and diagrams wherever necessary to support your explanation.
- (Q10.) Compare the layer structure and parameter count of VGG and ResNet in detail. Discuss how their architectural differences impact model complexity, performance, and training efficiency.

NOTE: The deadline for submitting the assignment is 30th May 2025.